Team Charter (M1)

Project name: MedMind — medication reminders and medication status for the caregivers of older adults

GitHub repo: https://github.com/yaroo393-png/MedMind

Team Members

Student ID	Name	GitHub	Role	Feature owned
6931503064	Yar Oo	yaroo393-png	Project Manager	FR-02 label scanning + all project documents
6931503035	Takkan Khanlui	V31311	System Designer	FR-01 pairing, consent and data access rules
6931503066	Rossatorn Sangkaew	IDK-SleepyBro	UX/UI Designer	FR-03 meal reminders and reminder screen
6931503049	Nichanan Lalua	KaidangSuk	Prototype Developer	FR-04 confirming a meal and home screen
6931503079	Suchitra Khomdee	Khaiwann	QA / Tester	FR-05 caregiver alert, daily status and test plan

PROBLEM

The problem, in 2–3 sentences: Older adults on fixed-time tablets or capsules may forget a dose, take it late, or not remember whether they took it, while the child who cares for them is at work and cannot follow up. Alarms, pill boxes, calls and LINE messages cannot confirm the status and report it to the caregiver automatically.

Who exactly has it: (1) older adults in Thailand on fixed-time tablets or capsules who use a smartphone and whose child or relative sets things up for them; (2) the adult child / caregiver who is not with them during the day.

Why it matters: a missed or mistimed dose means the treatment is not taken as prescribed, and the caregiver has to call or send a LINE message to check on every meal.

Scope: in — FR-01 to FR-05 (the MVP); FR-09 AI weekly summary only after the MVP. Out — reminders while the page is closed or the screen is locked, as-needed medicines, liquids and half tablets, reading handwriting, any medical advice.

Success measure: the caregiver is contacted only about meals that are not confirmed, instead of calling or messaging for every meal, and every meal not confirmed within 15 minutes of its reminder reaches the caregiver as an alert.

GATE 1 — REACHABLE USERS

#	Name / role	How we reach them	When
1	Peerapap Kummongkon — older adult on daily medication	By phone	September 3, 2026, 12:00–12:20 PM
2	Mrs. Wun Daengsan — adult child / caregiver	By phone	September 4, 2026, 4:20–4:30 PM
3	Ms. Chanantida Nosuk — older adult on daily medication	By phone	September 4, 2026, 4:40–4:50 PM

GATE 2 — MVP SHAPE

FR	Planned functional requirement
FR-01	Pair the older adult and the caregiver with a QR code or 6-digit invite code (the older adult consents first; no account or password)
FR-02	Photograph a medication label → caregiver covers the name and HN → AI drafts the details → caregiver checks → save
FR-03	Reminder at each medication meal, with repeat reminders
FR-04	Confirm the whole meal with one tap
FR-05	Alert the caregiver when a meal is not confirmed + daily status of every meal

Which FR carries a real rule or decision? FR-03–05: each meal follows a timeline — reminders at T, T+5, T+10; not confirmed by T+15 → one caregiver alert; late confirmation until the next meal. FR-02: the reminder time comes from the label (“after a meal”, “N minutes before”, “at bedtime”) and is never guessed.

What data has to persist between sessions? kept in the shared online data store: the pairing (with consent), medicines (name, tablets per dose, meal / time, warnings), meal times and bedtime, and the confirmation record of each meal. Not stored: national ID, address, date of birth, phone number, hospital number, label photos.

GATE 3 — ACCESS

Data, licence or system we depend on	How we know we can get it	If it fails
Firebase Spark plan — Authentication (anonymous sign-in), Cloud Firestore (shared live data, offline queue), Hosting (HTTPS)	No payment method needed; no-cost quota of 50,000 Firestore reads and 20,000 writes a day (Firebase pricing page, checked 30 Oct 2026)	Hosting moves to GitHub Pages; usage checked daily against the quota
Firebase AI Logic + Gemini Developer API (free tier) — reads the covered label photo and returns only the medicine fields; later also writes the weekly summary (FR-09, Could)	Firebase AI Logic is free and the Gemini free tier needs no payment method, so the project stays on Spark (Firebase AI Logic pricing page, checked 31 Oct 2026); calls go through the Firebase SDK, so the API key stays on Google's side; free-tier content may be used to improve Google's products, so only images with the name, HN and other personal details covered are sent; 20 real labels tested before FR-02 is finished	Manual entry when offline or when the daily limit is reached
html5-qrcode (Apache-2.0) to scan, qrcode (MIT) to draw the QR code	Free open-source libraries; run in the browser	6-digit invite code
Real medicine labels for testing	From interviewees and team families, with consent; names and HN covered	Results reported as a small sample
Test phones: OPPO Reno 16 5G (ColorOS 16) as the older adult, a second team phone as the caregiver	Owned by the team	A laptop browser as the caregiver

PROCESS & TEAM

Chosen SDLC: Incremental — Increment 1: core loop (remind → confirm → alert, FR-03–05) on real shared data with a team test pair; Increment 2: pairing and AI label reading (FR-01, FR-02), because label-reading accuracy is the biggest unknown and must not block the core loop.